AGN X-ray Heating: Derivation of the Emissivity-to-Heating-Rate Pipeline

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This note derives, from the AGN spectral model down to the T_K evolution equation, every equation implemented in `ComputeTs.c/XRayHeatingFunctions.c` for the AGN X-ray channel, in parallel with the pre-existing stellar (GAL) pathway. Every symbol is cross-referenced to its code variable so this can be checked line-by-line against the source. The specific question under discussion why the soft and hard bands need distinct integration limits in the *luminosity-conversion* step but not, for the same reason, in `integrate_over_nu` is addressed explicitly in Section 4.1.

Contents

1 The AGN spectral model

Each band (soft, hard) is modelled as an independent power law in specific luminosity, pivoted at a threshold frequency $\nu_{\rm th}$ (`NuXrayThreshold`):

$$L_b(\nu) = L_{0,b} \left(\frac{\nu}{\nu_{\rm th}}\right)^{-\alpha_b}, \qquad b \in \{\text{soft}, \text{hard}\}, \tag{1}$$

where α_b is `SpecIndexXrayAGNSoft` or `SpecIndexXrayAGNHard`, and $L_{0,b}$ is the (as yet unknown) amplitude at the pivot. The two bands are defined over disjoint frequency ranges, split at the break frequency $\nu_{\rm break}$ (`NuXraySoftCut`):

$$\text{soft:}\quad \nu \in [\nu_{\rm th},\, \nu_{\rm break}], \tag{2}$$
$$\text{hard:}\quad \nu \in [\nu_{\rm break},\, \nu_{\rm max}], \qquad \nu_{\rm max} = \texttt{NuXrayMax}. \tag{3}$$

Each band's amplitude $L_{0,b}$ is fixed independently, by requiring that integrating Eq. (1) over that band's own frequency range reproduces a physically known, calibrated band luminosity $L_{{\rm band},b}$ (the accretion-derived L_X from the black-hole growth history, computed in `blackhole_feedback.c quasar_lx`, obscuration-corrected):

$$\int_{\nu_{{\rm lo},b}}^{\nu_{{\rm hi},b}} L_b(\nu)\, d\nu = L_{{\rm band},b}. \tag{4}$$

2 Step 1: the luminosity conversion factor

Define the *luminosity conversion factor* as the ratio of the pivot amplitude to the calibrated band luminosity,

$$C_b \equiv \frac{L_{0,b}}{L_{{\rm band},b}}, \tag{5}$$

which is exactly `Luminosity_converstion_factor_AGN_soft` / `_hard` in the code (up to the unit conversions in Section 2.3). Solving Eq. (4) for $L_{0,b}$ gives C_b.

2.1 Case $\alpha_b \neq 1$

$$L_{{\rm band},b} = L_{0,b}\, \nu_{\rm th}^{\alpha_b} \int_{\nu_{{\rm lo},b}}^{\nu_{{\rm hi},b}} \nu^{-\alpha_b}\, d\nu = L_{0,b}\, \nu_{\rm th}^{\alpha_b}\, \frac{\nu_{{\rm hi},b}^{1-\alpha_b} - \nu_{{\rm lo},b}^{1-\alpha_b}}{1-\alpha_b}, \tag{6}$$

$$\Rightarrow \quad C_b = \frac{(1-\alpha_b)\, \nu_{\rm th}^{-\alpha_b}}{\nu_{{\rm hi},b}^{1-\alpha_b} - \nu_{{\rm lo},b}^{1-\alpha_b}}. \tag{7}$$

Code check (soft band, `ComputeTs.c:1027-1037`):

```
Luminosity_converstion_factor_AGN_soft =
    pow(nu_hi, 1 - alpha) - pow(nu_lo, 1 - alpha);      // (nu_hi^(1-a) - nu_lo^(1-a))
Luminosity_converstion_factor_AGN_soft = 1.0 / ...;     // 1 / (...)
Luminosity_converstion_factor_AGN_soft *=
    pow(nu_th, -alpha) * (1 - alpha);                   // x nu_th^-a (1-a)
```

which is exactly Eq. (7), per band, with $(\nu_{{\rm lo},b}, \nu_{{\rm hi},b}) = (\nu_{\rm th}, \nu_{\rm break})$ for soft and $(\nu_{\rm break}, \nu_{\rm max})$ for hard.

2.2 Case $\alpha_b = 1$ (the singular case)

When $\alpha_b \to 1$, $1-\alpha_b \to 0$ and Eq. (7) is 0/0: the power-law primitive $\nu^{1-\alpha}/(1-\alpha)$ degenerates, because $\int \nu^{-1} d\nu = \ln \nu$ instead. Re-deriving directly for $\alpha_b = 1$:

$$L_{{\rm band},b} = L_{0,b}\, \nu_{\rm th} \int_{\nu_{{\rm lo},b}}^{\nu_{{\rm hi},b}} \nu^{-1}\, d\nu = L_{0,b}\, \nu_{\rm th}\, \ln\!\left(\frac{\nu_{{\rm hi},b}}{\nu_{{\rm lo},b}}\right), \tag{8}$$

$$\Rightarrow \quad C_b = \frac{1}{\nu_{\rm th}\, \ln(\nu_{{\rm hi},b}/\nu_{{\rm lo},b})}. \tag{9}$$

Code check (`ComputeTs.c:1021-1026`):

```
if (fabs(SpecIndexXrayAGNSoft - 1.0) < REL_TOL) {
    Luminosity_converstion_factor_AGN_soft =
        (NuXrayThreshold*NU_over_EV) * log(NuXraySoftCut / NuXrayThreshold);
    Luminosity_converstion_factor_AGN_soft = 1.0 / ...;
}
```

which is exactly Eq. (9). This confirms the branch is not an arbitrary numerical safeguard: $\alpha_b = 1$ is a genuine mathematical singularity of the power-law primitive (a log-divergence of the naive formula), not just a coding edge case, so it needs its own closed-form solution rather than a tolerance-based patch of Eq. (7). The same structure (generic $\alpha \neq 1$ branch + $\alpha = 1$ log branch) already exists for GAL and, under `USE_MINI_HALOS`, for PopIII the AGN soft/hard code simply repeats it per band.

2.3 Unit conversion

C_b as derived above is dimensionless (it is a pure ratio of frequency-integrated shape to the same shape's own normalization). The code additionally divides by `PLANCK` (Section `ComputeTs.c:1042,1066`): this converts a specific-luminosity amplitude into a specific *photon*-luminosity amplitude ($L_\nu/h\nu_{\rm th}$-type normalization), consistent with everything downstream (`integrate_over_nu`'s integrand, Section 4) being expressed per unit photon energy. Unlike the GAL/PopIII conversion factors, no `SEC_PER_YEAR` factor is applied: GAL/III start from a star-formation-rate calibration ($M_\odot\,{\rm yr}^{-1}$) and need that factor to convert to a per-second rate; the AGN band luminosities are already luminosities (${\rm erg\,s^{-1}}$), so no such conversion is needed.

3 Step 2: the redshift-dependent prefactor

The pivot amplitude, once fixed by C_b, still needs to be turned into the proper prefactor multiplying the frequency-shape integral at each emission redshift z' (`zp`). This is `const_zp_prefactor_AGN_soft/_hard`:

$$\mathcal{A}_b(z') = \frac{C_b}{\nu_{\rm th}}\, c\,(1+z')^{\alpha_b+3}. \tag{10}$$

Code check (`ComputeTs.c:1117-1131`):

```
const_zp_prefactor_AGN_soft = Luminosity_converstion_factor_AGN_soft
    / (NuXrayThreshold * NU_over_EV) * SPEED_OF_LIGHT
    * pow(1 + zp, SpecIndexXrayAGNSoft + 3);
```

which is Eq. (10) directly. This functional form $(1+z')^{\alpha+3}$ is inherited unmodified from the pre-existing GAL prefactor (itself the standard 21cmFAST/Ts.c form of Pritchard & Furlanetto 2007 / Mesinger, Furlanetto & Cen 2011): the +3 comes from proper number-density dilution $\propto (1+z')^3$, and the α_b comes from redshifting a $\nu^{-\alpha_b}$ spectrum (each photon's frequency, and hence its place on the power law, redshifts along with z'). The AGN channel does not re-derive this it reuses the identical cosmological scaling, per band, with that band's own α_b and C_b.

Both prefactors are checked for finiteness immediately (`ComputeTs.c:1121,1132`) and the run aborts rather than silently propagating a bad value this guards against a zero-width band ($\nu_{{\rm hi},b} = \nu_{{\rm lo},b}$, e.g. `NuXraySoftCut==NuXrayThreshold`), which would make C_b divide by zero.

4 Step 3: the frequency-shape integral (`integrate_over_nu`)

This is a *different* integral, doing a *different* job. Where C_b (Section 2) fixes one number the overall amplitude of the band, so that the model matches a known calibration luminosity `integrate_over_nu` computes, at every radial shell R (i.e. every look-back shell z'', `zpp`) and every ionization state x_e, the deposition-efficiency-weighted photon-number shape kernel:

$$\mathcal{K}_b(z', z'', x_e) = \int_{\nu_{\rm lo}(z',z'')}^{\nu_{\rm hi}(z',z'')} f_{\rm dep}(\nu, x_e) \left(\frac{\nu}{\nu_{\rm th}}\right)^{-\alpha_b - 1} d\nu, \tag{11}$$

where $f_{\rm dep}$ is the frequency- and ionization-dependent deposition/ionization/Lyα-production efficiency (`interp_fheat`, etc. inside `integrand_in_nu_heat_integral` and its ionization/Lyα siblings) and the extra power of ν^{-1} relative to Eq. (1) converts specific *luminosity* into specific *photon-number* flux ($L_\nu/h\nu$). This kernel is a pure numerical integral over whatever limits are handed to it it performs no calibration and needs no conversion factor, because nothing about it is required to reproduce an independently-known total.

4.1 Why its soft/hard limits differ from C_b's

Both places use different limits for soft vs. hard, but the two splits answer two different physical questions, which is the direct answer to "why are they needed here but not for `integrate_over_nu`":

- **C_b's limits** ($[\nu_{\rm th}, \nu_{\rm break}]$ vs. $[\nu_{\rm break}, \nu_{\rm max}]$, fixed, rest-frame) exist because *calibration is band-specific by definition*: C_b is only meaningful once "the band" is fixed, since it is derived by demanding $\int_{\rm band} L_b(\nu)\, d\nu$ equals that band's own known luminosity (Eq. (4)). Soft and hard are calibrated against physically distinct observables (soft-band and hard-band L_X, with distinct obscuration corrections and, potentially, distinct α_b), so they necessarily use distinct, fixed limits. This step happens once per redshift step, with no z'' dependence at all.

- **`integrate_over_nu`'s limits** (`lower/upper_int_limit_AGN_soft/hard`, recomputed every (z', z'') pair) exist for an unrelated reason: they track *which part of the rest-frame spectrum can physically still be soft or hard by the time it arrives at z'*. A photon emitted at z'' in the rest-frame hard band redshifts by $(1+z')/(1+z'')$ on its way to z', and can redshift down into what is, at z', the soft band hence the

  ```
  * (1. + zp) / (1. + zpp)
  ```

  factor on every AGN soft/hard limit (`ComputeTs.c:793-821`), absent from GAL/III's limits because GAL/III use one single band with no internal soft/hard split to redshift across. On top of that redshift scaling, the lower edge of each band is further clamped to $\max(\cdot,\, \nu_{\tau=1}(z', z''))$, the frequency at which the IGM becomes optically thick along that shell (`nu_tau_one_agn`) a purely radiative-transfer constraint with no analogue in the calibration step at all.

In short: C_b's limits define *what the band is*, once, in the rest frame, for calibration purposes. `integrate_over_nu`'s limits define *what fraction of each band is currently visible*, recomputed every shell, for radiative-transfer purposes. They are both "different limits for soft vs. hard" but answer unrelated questions one is about the definition of a band, the other about a shifting horizon within it.

5 Step 4: assembling the heating-rate ODE term

Per shell $z''_{\rm ct}$ (in `evolveInt`, `XRayHeatingFunctions.c:1409-1423`):

$$I_b(z'', x_e) = X_b(z'')\,(1+z'')^{-\alpha_b}\,\mathcal{K}_b(z', z'', x_e), \tag{12}$$

$$\left.\frac{dx_{\rm heat}}{dt}\right|_b = \mathcal{A}_b(z') \sum_{z''} \frac{dt}{dz''}\,\Delta z''\, I_b(z'', x_e), \tag{13}$$

where $X_b(z'')$ is `XAGN_soft/_hard`[z"], the AGN band emissivity for that shell (an `SMOOTHED_AGN_(soft/hard)` grid value, itself built from `gal->BHXrayEmissivity` in `blackhole_feedback.c`). The $(1+z'')^{-\alpha_b}$ factor here is the emission-side counterpart of the $(1+z')^{\alpha_b+3}$ in $\mathcal{A}_b(z')$ (Eq. (10)) together they implement the same $(1+z'/1+z'')^{\alpha_b}$ spectral redshifting used to shift the `integrate_over_nu` limits in Section 4.1, applied this time to the amplitude rather than to the integration bounds. dt/dz'' is `dtdz(zpp)`, $\Delta z''$ is the shell width (`dzpp`), and the sum runs over `TsNumFilterSteps` radial/temporal shells.

Soft and hard are summed only after each has its own $\mathcal{A}_b$ applied (`XRayHeatingFunctions.c:1457-1469`):

$$\left.\frac{dx_{\rm heat}}{dt}\right|_{\rm AGN} = \left.\frac{dx_{\rm heat}}{dt}\right|_{\rm soft} + \left.\frac{dx_{\rm heat}}{dt}\right|_{\rm hard}, \tag{14}$$

and this combined AGN heating rate enters dT_K/dz' (`deriv[3] / dansdz[3]`) exactly where the stellar term does, alongside adiabatic cooling and Compton heating (unchanged from the pre-existing GAL/III pathway).

6 Symbol / code cross-reference

Symbol	Meaning	Code name
$\nu_{\rm th}$	pivot frequency	`NuXrayThreshold`
$\nu_{\rm break}$	soft/hard split	`NuXraySoftCut`
$\nu_{\rm max}$	hard band ceiling	`NuXrayMax`
α_b	band spectral index	`SpecIndexXrayAGNSoft / Hard`
$L_{{\rm band},b}$	calibrated band luminosity	`quasar_lx` (obscuration-corrected)
C_b	luminosity conversion factor	`Luminosity_converstion_factor_AGN_soft/hard`
$\mathcal{A}_b(z')$	redshift prefactor	`const_zp_prefactor_AGN_soft/hard`
$\mathcal{K}_b(z', z'', x_e)$	frequency shape kernel	`freq_int_heat_tbl_AGN_soft/hard`
$X_b(z'')$	per-shell AGN emissivity	`XAGN_soft/hard[R_ct]`
$\nu_{\tau=1}(z', z'')$	opacity horizon	`nu_tau_one_agn`

7 Open items for the meeting

- Confirm the sign/exponent convention in Eq. (1) ($L_\nu \propto \nu^{-\alpha}$) matches the intended definition of `SpecIndexXrayAGNSoft/Hard` as used elsewhere in the paper/thesis.
- Confirm whether $L_{{\rm band},b}$ (the calibration target) should itself already be obscuration-corrected at the point it enters Eq. (4), or whether obscuration is meant to apply only downstream (via `NHfrac/NHTrans`) this derivation assumes the latter (obscuration is applied separately, not baked into C_b).

- Verify the $\alpha_b = 1$ branch's tolerance (`REL_TOL` $= 10^{-5}$, `meraxes.h:65`) is tight enough not to introduce a discontinuity in C_b near $\alpha_b = 1$ for realistic parameter sweeps.